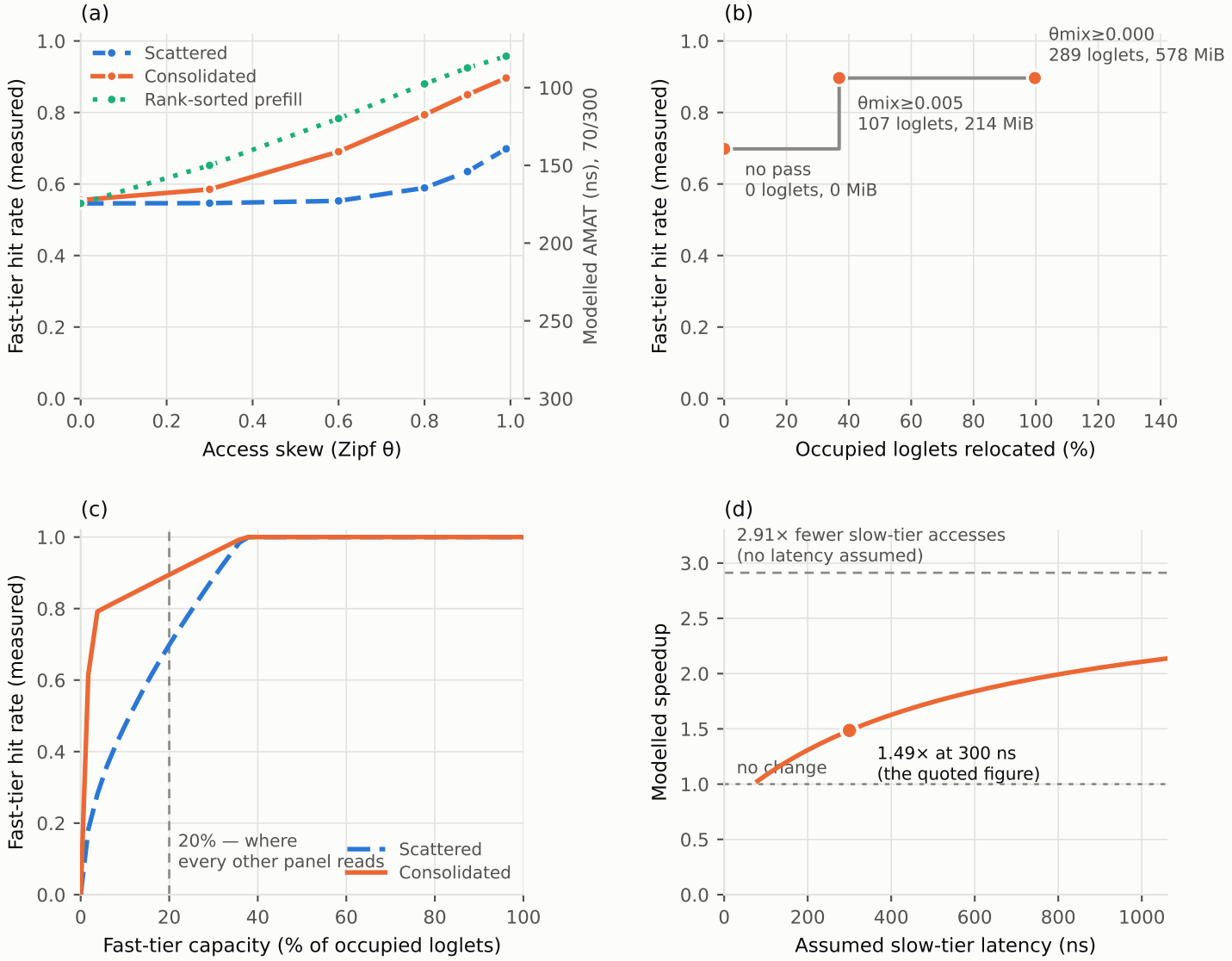
(a)
Scattered
Consolidated
Rank-sorted prefill
Fast-tier hit rate (measured)
Modelled AMAT (ns), 70/300
Access skew (Zipf θ)
(b)
θmix≥0.000
289 loglets, 578 MiB
θmix≥0.005
107 loglets, 214 MiB
no pass
0 loglets, 0 MiB
Fast-tier hit rate (measured)
Occupied loglets relocated (%)
(c)
20% — where
every other panel reads
Scattered
Consolidated
Fast-tier hit rate (measured)
Fast-tier capacity (% of occupied loglets)
(d)
2.91× fewer slow-tier accesses
(no latency assumed)
no change
1.49× at 300 ns
(the quoted figure)
Modelled speedup
Assumed slow-tier latency (ns)